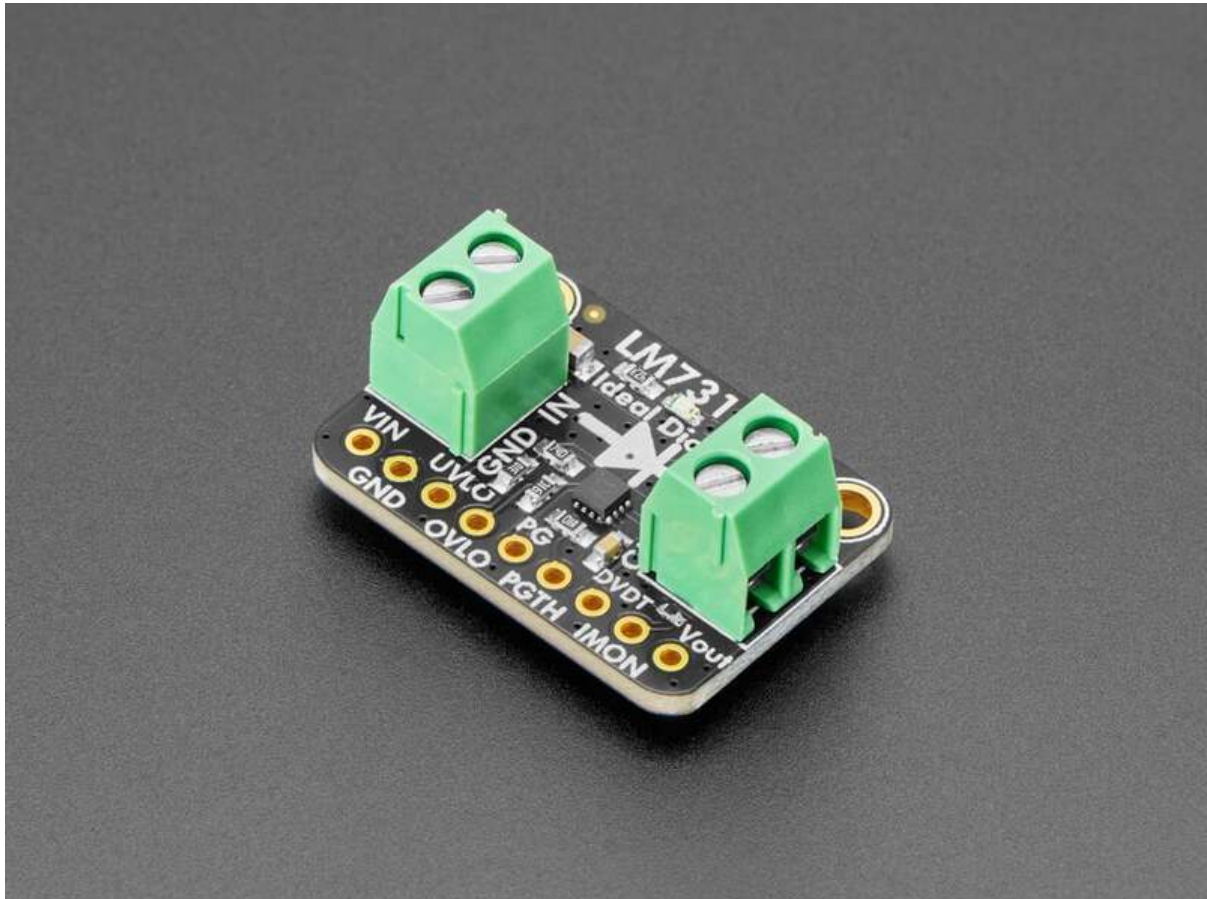




Adafruit LM73100 Ideal Diode Breakout

Created by Liz Clark



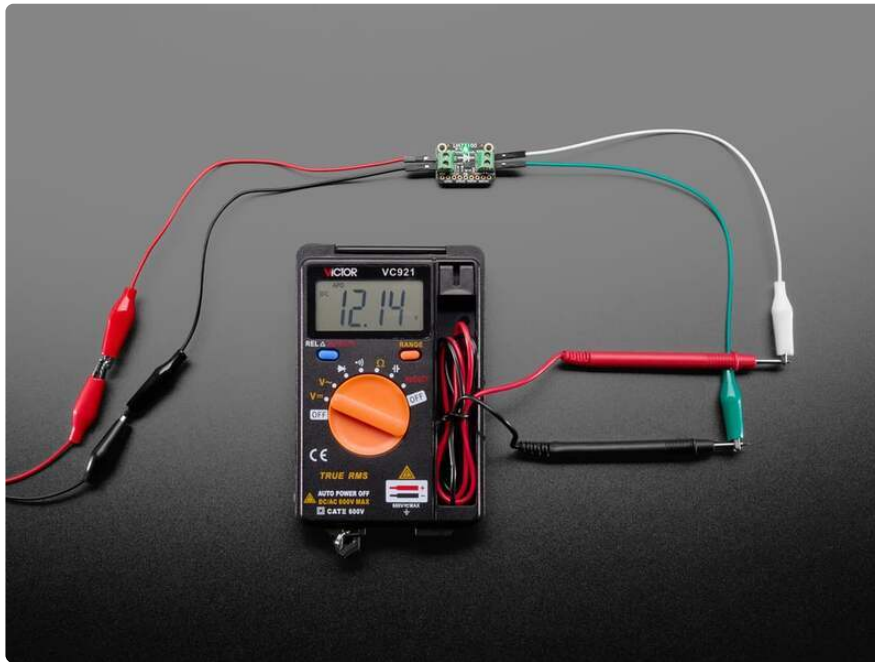
<https://learn.adafruit.com/adafruit-lm73100-ideal-diode-breakout>

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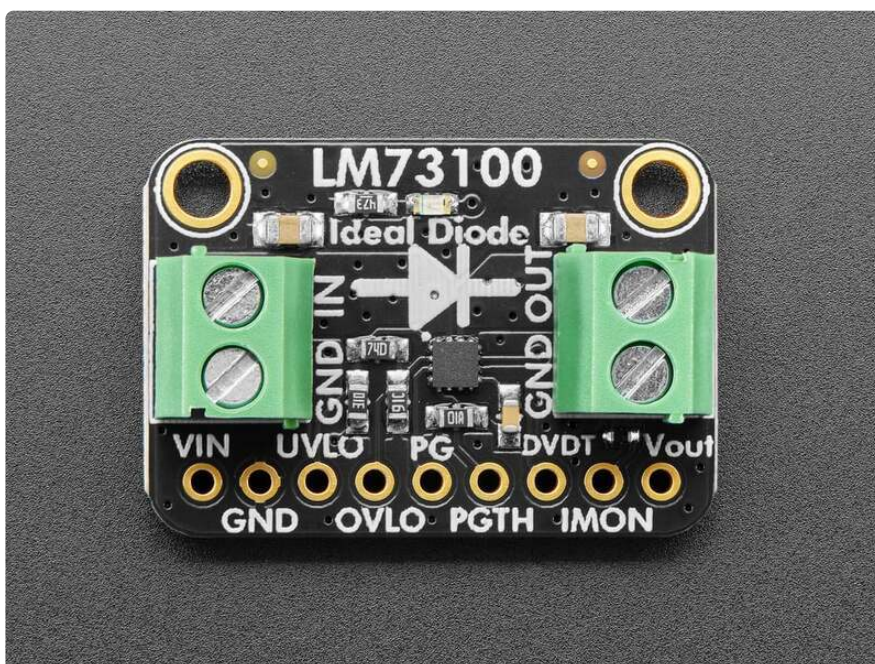
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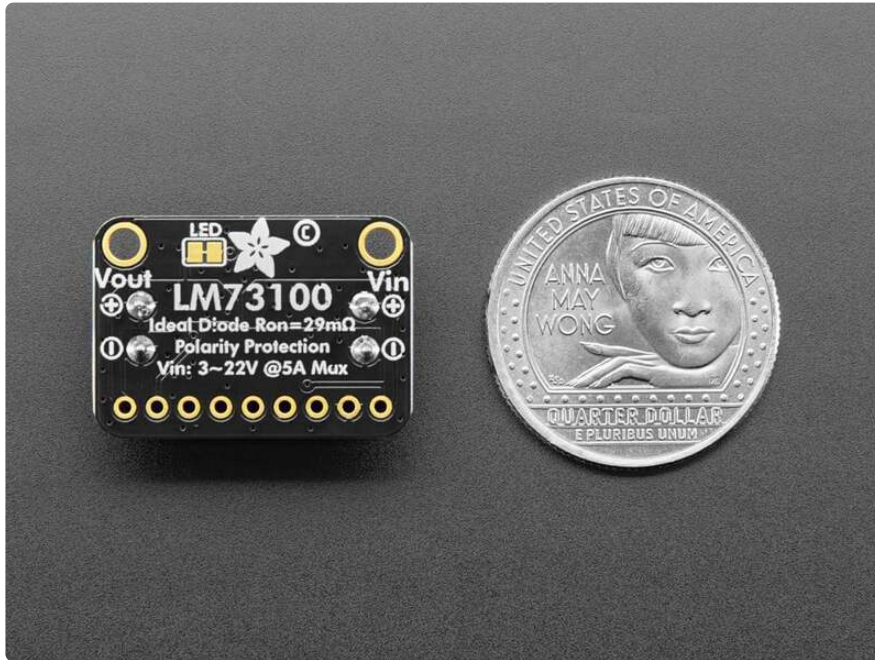
Overview



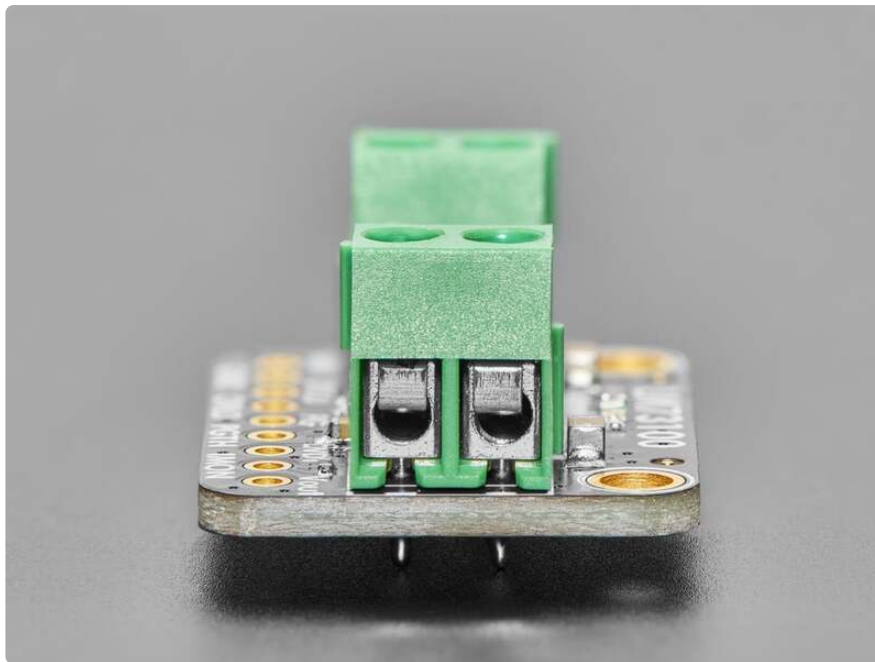
If your project is moving a lot of power around, and efficiency is key, an **Adafruit LM73100 Ideal Diode breakout** is a great way to keep your power budget tidy! As long as your power supply is 3 to 23V DC and up to 5A, use this in place of a diode to keep current flowing in only one direction. However, unlike a diode, you won't suffer a 1V+ forward drop which just gets lost as heat. Instead, the internal chip has two back-to-back FETs, with a typical $R_{ds(on)}$ of 30 m Ω . That means that even with 5A, you'll lose around 150mV. This makes it perfect for efficiency-sensitive setups like solar panels and LiPoly batteries.



Unlike our [LM62200 dual ideal diode](http://adafru.it/5830) (<http://adafru.it/5830>), this board acts as 'one' diode only - but it can handle much higher voltages and currents! There are also some niceties we added like solder-free operation thanks to on-board terminal blocks, and a green LED that will light when power is going in the right direction. You can also monitor the current draw by reading the analog voltage off of IMON, handy if you don't want a separate power monitoring kit.

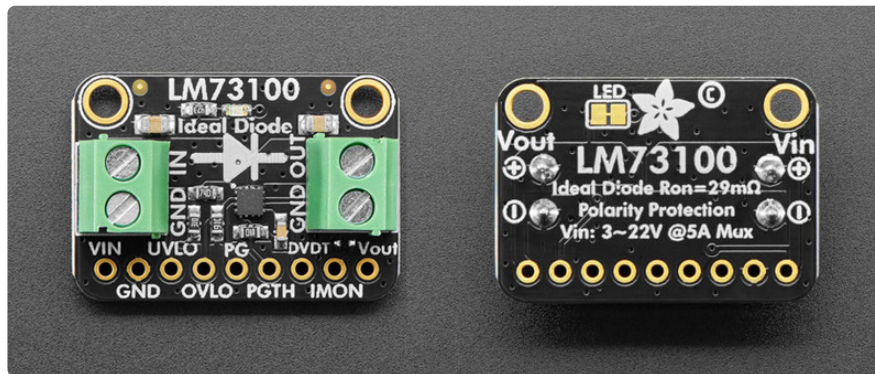


To make it breadboard friendly, we've placed it on a small PCB with 9 x 0.1" breakout pads. We include a bit of header. Some light soldering is required to attach the header for breadboarding use. Or you can use direct wiring to the terminal blocks if you want to insert it directly into a circuit without a breadboard.



- Wide operating input voltage range: 2.7 V to 23 V
 - 28 V absolute maximum
 - Withstands negative voltages up to -15 V
- Integrated back-to-back FETs with low OnResistance: $R_{ON} = 28.4 \text{ m}\Omega$ (typ)
- Ideal diode operation with true reverse current blocking
- Fast overvoltage protection
- 1.2- μs (typ) response time
- Fast-trip response to transient overcurrents during steady state
 - 500-ns (typ) response time
 - Latch-off after fault
- Analog load current monitor output (IMON)
 - Current range: 0.5 A to 5.5 A
 - Accuracy: $\pm 15\%$ (max) ($I_{OUT} \geq 1 \text{ A}$)
- Overtemperature protection
- Power Good indication (PG)

Pinouts



Power

- **VIN / IN (terminal block)** - To power the board, give it 3V to 22V DC. The power input pin is broken out at the bottom edge of the board (**VIN**) and on the terminal block on the left side of the board (**IN**).
- **Vout / OUT (terminal block)** - this is the output from the ideal diode. It is broken out at the bottom edge of the board (Vout) and on the terminal block on the right side of the board (**OUT**).
- **GND / GND (terminal block)** - common ground. The ground pin is broken out at the bottom edge of the board (**GND**) and on the terminal blocks on both sides of the board (GND).

LM73100 Pins

- **UVLO** - This is the active high enable pin. It is also used to set the undervoltage lockout threshold. At 3V, that is 1.2V and at 23V that is 9.2V.
- **OVLO** - This is the overvoltage lockout threshold pin. At 3V, that is 0.15V and at 23V it is 1.2V.
- **PG** - This is the power good indication pin. It is high when a proper power supply is detected.
- **PGTH** - This is the power good threshold pin. It can be set with a threshold setting to check against for the power good pin.
- **DVDT** - This is the slew rate/inrush current control pin.
- **IMON** - This is the current monitoring pin. You can read the analog voltage from this pin as a current monitor.

Power LED and Jumper

- **Power LED** - In the top center of the board, below the LM73100 label on the board silk, is the power LED, It is a green LED.

- **LED jumper** - This jumper is located on the back of the board and is labeled **LED** on the board silk. Cut the trace on this jumper to cut power to the power LED.

CircuitPython

This code will not run on a Raspberry Pi single board computer since Pi SBCs do not have an analog input.

It's easy to use the **LM73100 Ideal Diode** with CircuitPython, and the [analogio core module](https://adafru.it/1anp) (<https://adafru.it/1anp>). This module allows you to easily write Python code to read analog input data from the LM73100 IMON pin.



Lithium Ion Battery Pack - 3.7V 4400mAh

Need a big battery for your project? This lithium-ion pack is made of 2 balanced 2200mAh cells for a total of 4400mA capacity! The cells are connected in parallel and spot-welded to a...

<https://www.adafruit.com/product/354>



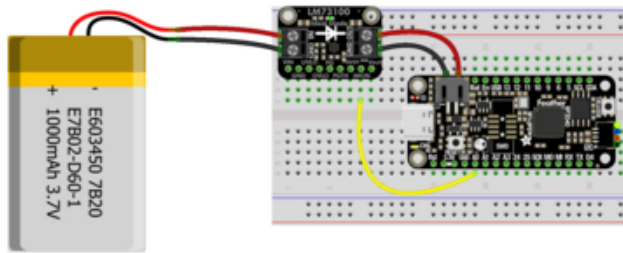
JST-PH Battery Extension Cable - 500mm

By popular demand, we now have a handy extension cord for all of our JST PH-terminated battery packs (such as our Lilon/LiPoly and 3xAAA holders). One end has a JST-PH compatible...

<https://www.adafruit.com/product/1131>

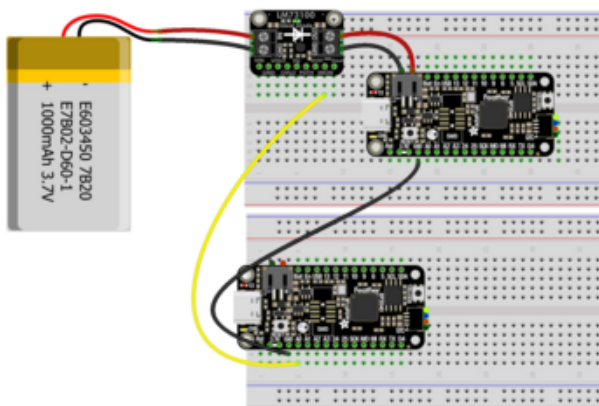
CircuitPython Microcontroller Wiring

First wire up the breakout to your board exactly as follows. The following is the breakout wired to a Feather RP2040 with the breakout inline with a lipo battery:



- LiPo battery positive to ideal diode VIN (red wire)
- LiPo battery negative to ideal diode GND (black wire)
- Ideal diode OUT to Feather JST-PH positive (red wire)
- Ideal diode GND to Feather JST-PH negative (black wire)
- Ideal diode IMON to Feather pin A0 (yellow wire)

Note that when you connect the Feather via USB for serial monitoring, the battery goes into charge mode and as a result isn't drawing any current through the battery. You can use an additional Feather or microcontroller to get current readings for this demo:



- LiPo battery positive to ideal diode VIN (red wire)
- LiPo battery negative to ideal diode GND (black wire)
- Ideal diode OUT to Feather 1 JST-PH positive (red wire)
- Ideal diode GND to Feather 1 JST-PH negative (black wire)
- Feather 1 GND to Feather 2 GND (black wire)
- Ideal diode IMON to Feather 2 pin A0 (yellow wire)

In practice, for a battery powered IoT project for example, you would likely be reading the current from the battery connected to the target Feather in code or logging wirelessly.

CircuitPython Usage

To use with CircuitPython, you need to update **code.py** with the example script.

Thankfully, we can do this in one go. In the example below, click the **Download Project Bundle** button below to download the necessary libraries and the **code.py** file

in a zip file. Extract the contents of the zip file, and copy the **code.py** file to your **CIRCUITPY** drive.

Your **CIRCUITPY/lib** folder will be empty since the example is only using core modules that are built into the **CIRCUITPY** drive.



The **/lib** folder will be empty since the example is only using core modules.

Example Code

Once everything is saved to the **CIRCUITPY** drive, [connect to the serial console \(https://adafru.it/Bec\)](https://adafru.it/Bec) to see the data printed out!

```
# SPDX-FileCopyrightText: 2025 Liz Clark for Adafruit Industries
#
# SPDX-License-Identifier: MIT
# LM73100 IMON Current Monitoring

import time
import board
from analogio import AnalogIn

RIMON = 1500.0 # 1.5kΩ
GIMON = 2.5 # μA/A
analog_in = AnalogIn(board.A0)

def get_voltage(pin):
    return (pin.value * 3.3) / 65536

print("LM73100 Current Monitor Started")
print("=====")
print(f"RIMON: {RIMON} ohms")
print(f"GIMON: {GIMON} μA/A")
print("=====\\n")

while True:
    # Read voltage from IMON pin
    vimon = get_voltage(analog_in)

    # Calculate output current
    iout_A = vimon / (RIMON * GIMON)
    iout_mA = iout_A * 1000.0
```

```
print(f"ADC: {analog_in.value} | VIMON: {vimon:.3f}V | Current: {iout_mA:.2f}
mA")

time.sleep(0.5)
```

```
ADC: 55405 | VIMON: 2.914V | Current: 0.78 mA
ADC: 37593 | VIMON: 2.019V | Current: 0.54 mA
ADC: 20725 | VIMON: 1.136V | Current: 0.30 mA
ADC: 0 | VIMON: 0.000V | Current: 0.00 mA
ADC: 0 | VIMON: 0.000V | Current: 0.00 mA
ADC: 64 | VIMON: 0.000V | Current: 0.00 mA
ADC: 1472 | VIMON: 0.035V | Current: 0.01 mA
ADC: 18868 | VIMON: 0.868V | Current: 0.23 mA
ADC: 40953 | VIMON: 2.037V | Current: 0.54 mA
ADC: 56701 | VIMON: 2.953V | Current: 0.79 mA
ADC: 54109 | VIMON: 2.865V | Current: 0.76 mA
ADC: 36520 | VIMON: 1.948V | Current: 0.52 mA
ADC: 12995 | VIMON: 0.728V | Current: 0.19 mA
ADC: 0 | VIMON: 0.000V | Current: 0.00 mA
ADC: 0 | VIMON: 0.000V | Current: 0.00 mA
```

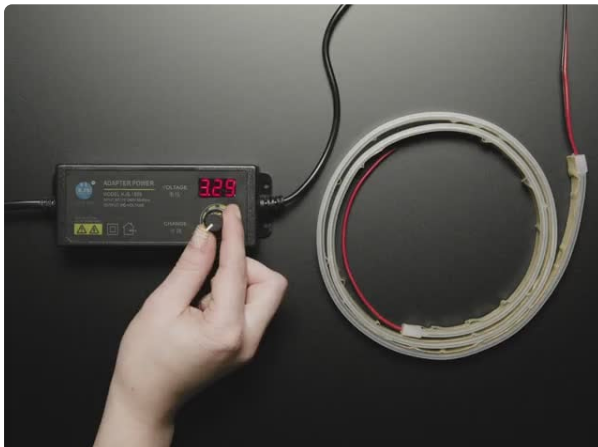
You'll see the raw analog reading, the **IMON** voltage reading and the current draw from the **IMON** pin printed out to the serial console. The code uses the formula from the datasheet (page 20, section 7.3.5 "Analog Load Current Monitor Output"):

$$I_{out} (A) = (V_{imon}(V) \times 10^{-6}) / (R_{imon}(\Omega) \times G_{imon}(\mu A/A))$$

to calculate the current from the voltage output from the **IMON** pin as an analog reading.

Arduino

Using the LM73100 Ideal Diode with Arduino involves wiring up the breakout to your Arduino-compatible microcontroller and running the provided example code.



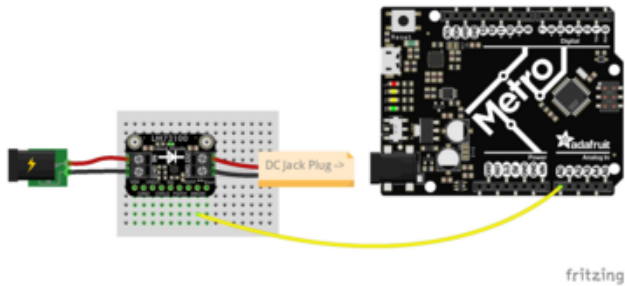
[Adjustable Power Supply with 2.1mm / 5.5mm DC - 3V to 12V at 5A](https://www.adafruit.com/product/4880)

Put your Snap! cassette on and sing along to "I got the power!" with this super useful power supply adapter where...

<https://www.adafruit.com/product/4880>

Wiring

Wire as shown for a **5V** board like an Uno, with the breakout inline with an external DC power supply. If you are using a **3V** board, like an Adafruit Feather, wire the board's 3V pin to the breakout VIN.



Power supply positive to ideal diode VIN (red wire)

Power supply negative to ideal diode GND (black wire)

Ideal diode OUT to DC jack positive (red wire)

Ideal diode GND to DC jack negative (black wire)

DC jack output to Metro DC jack

Ideal diode IMON to Metro pin A0 (yellow wire)

No additional libraries are needed for this example.

Example Code

```
// SPDX-FileCopyrightText: 2025 Liz Clark for Adafruit Industries
//
// SPDX-License-Identifier: MIT

// LM73100 IMON Current Monitoring

const int IMON_PIN = A0;           // IMON connected to analog pin A0
const float RIMON = 1500.0;        // RIMON resistor value in ohms (1.5kΩ)
const float GIMON = 2.5;           // GIMON typical value in μA/A (from datasheet,
typical is 2.5)
const float VREF = 5.0;            // Arduino reference voltage (5V for most
Arduinos)
const int ADC_RESOLUTION = 1024;

void setup() {
  Serial.begin(9600);
  pinMode(IMON_PIN, INPUT);

  Serial.println("LM73100 Current Monitor Started");
  Serial.println("=====");
  Serial.print("RIMON: ");
  Serial.print(RIMON);
  Serial.println(" ohms");
  Serial.print("GIMON: ");
  Serial.print(GIMON);
```

```

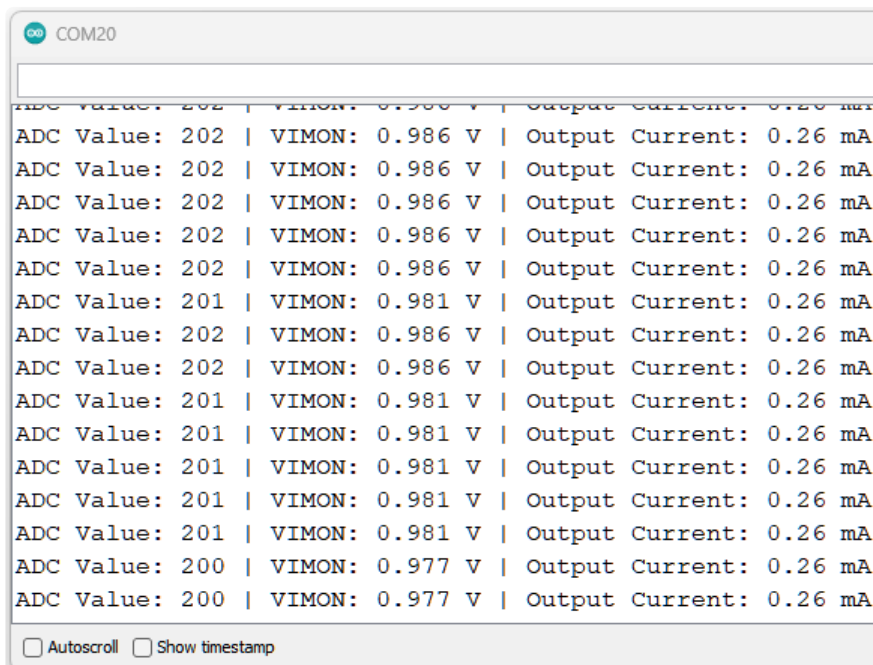
Serial.println(" μA/A");
Serial.println("=====\n");
}

void loop() {
  int adcValue = analogRead(IMON_PIN);
  float vimon = (adcValue * VREF) / ADC_RESOLUTION;
  float iout_A = vimon / (RIMON * GIMON);
  float iout_mA = iout_A * 1000.0;

  Serial.print("ADC Value: ");
  Serial.print(adcValue);
  Serial.print(" | VIMON: ");
  Serial.print(vimon, 3);
  Serial.print(" V | Output Current: ");
  Serial.print(iout_mA, 2);
  Serial.print(" mA (");
  Serial.print(iout_A, 3);
  Serial.println(" A)");

  delay(500); // Read every 500ms
}

```



Upload the sketch to your board and open up the Serial Monitor (**Tools -> Serial Monitor**) at 115200 baud. You'll see the raw analog reading, the **IMON** voltage reading and the current draw from the **IMON** pin printed out to the Serial Monitor. The code uses the formula from the datasheet (page 20, section 7.3.5 "Analog Load Current Monitor Output"):

$$I_{out} (A) = (V_{imon}(V) \times 10^{-6}) / (R_{imon}(\Omega) \times G_{imon}(\mu A/A))$$

to calculate the current from the voltage output from the **IMON** pin as an analog reading.

Downloads

Files

- [LM73100 Datasheet \(https://adafru.it/1anB\)](https://adafru.it/1anB)
- [EagleCAD PCB files on GitHub \(https://adafru.it/1anC\)](https://adafru.it/1anC)
- [3D models on GitHub \(https://adafru.it/1anD\)](https://adafru.it/1anD)
- [Fritzing object in the Adafruit Fritzing Library \(https://adafru.it/1anE\)](https://adafru.it/1anE)

Schematic and Fab Print

